

English
Medium



EDU TERIA

72nd BPSC

Prelims Online Batch

PHYSICS

LECTURE

05

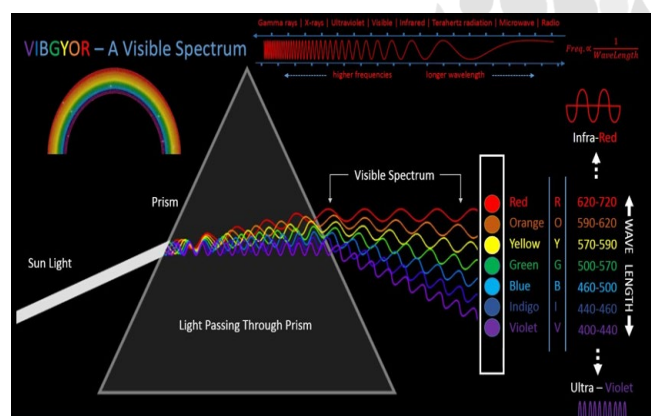
Light – Part - 04



Light – Part - 04

Dispersion of light

- Dispersion of light is the splitting of white light into its constituent colors (VIBGYOR) as it passes through a medium like a prism.
- This phenomenon occurs because the refractive index of the medium is different for each color, causing them to bend at different angles, violet light bends the most, and red light bends the least.

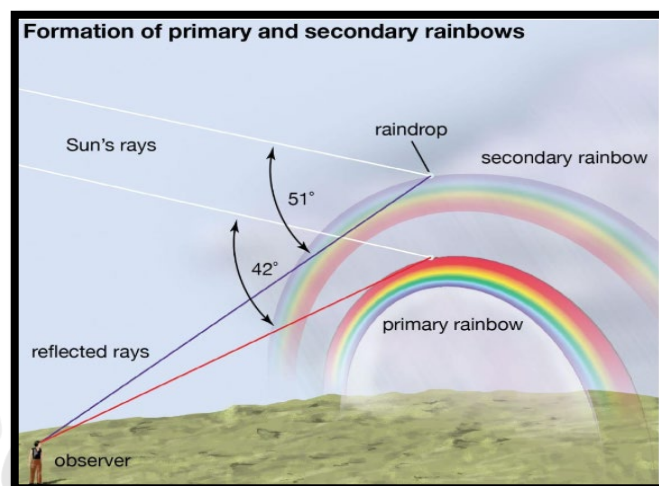


- After rainfall water droplets are present in atmosphere act as prism.



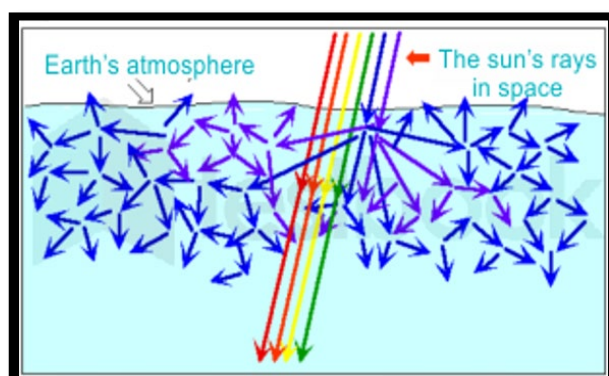
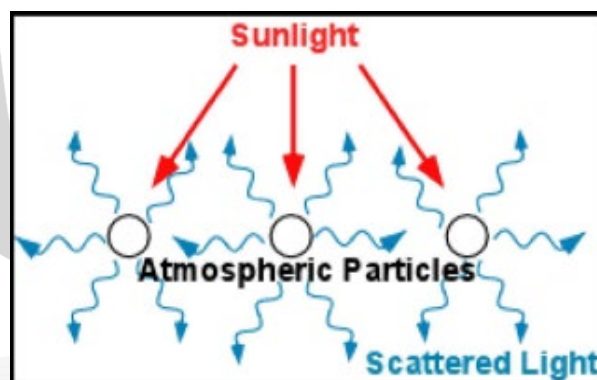
Types of Rainbows

Primary Rainbow	Secondary Rainbow
Makes angle of 42° with human eye	Makes 52° Angle with human eye.
Inner violet colour outer band Red colour	Inner band with red colour outer band with violet colour.



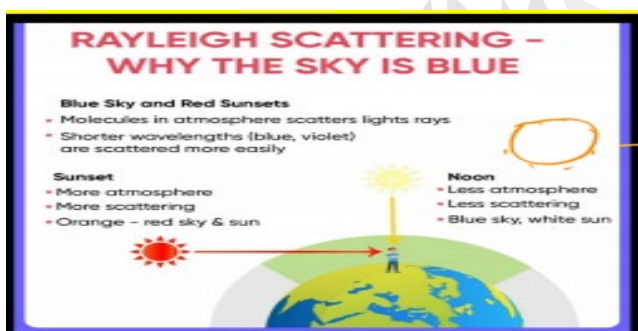
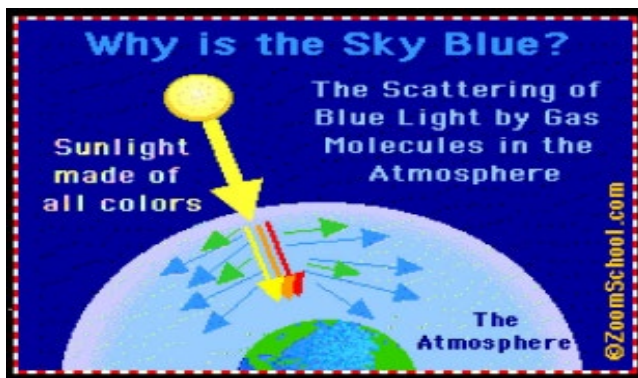
Scattering of light

- Scattering of light is the phenomenon where a beam of light is redirected in various directions after interacting with tiny particles, such as air molecules or dust.
- When light strikes these particles, it is absorbed and then re-emitted in different directions.
- The strength of scattering depends on the wavelength of light and the size of the particles, with shorter wavelengths scattering more easily.





for effects like the colors on soap bubbles or oil slicks.



Scattering of Light

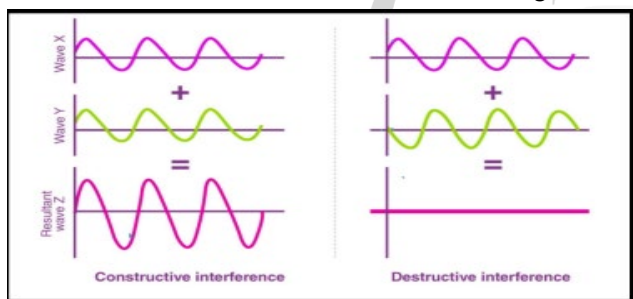
- Explains blue colour of sky
- Red colour of sunrise & sunset
- Rayleigh Scattering Law

$$\text{Intensity} \propto \frac{1}{\lambda^4}$$

Interference of Light

Interference of light is a phenomenon of Wave Optics in which two or more coherent light waves superpose (overlap) and produce a pattern of bright and dark fringes.

- Constructive interference → Bright fringes
- Destructive interference → Dark fringes



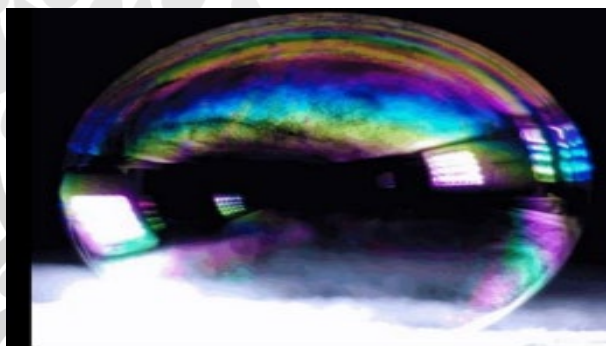
- Young's double-slit experiment proves wave nature of light
- It causes patterns of bright and dark bands called interference fringes and is responsible

Conditions for Interference

- For sustained interference pattern:
- Sources must be coherent (same frequency & constant phase difference)
- Light should be monochromatic (single wavelength)
- Waves should have comparable amplitudes
- Sources should be close enough

Example of Interference of Light

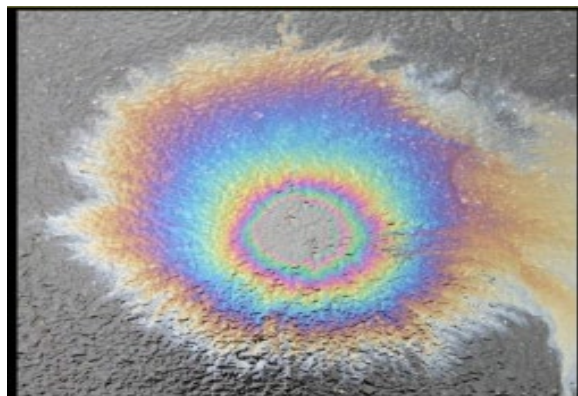
1. Color appearance in soap bubble.



2. Color in CD-DVD surface



3. Colorful oil film on water surface



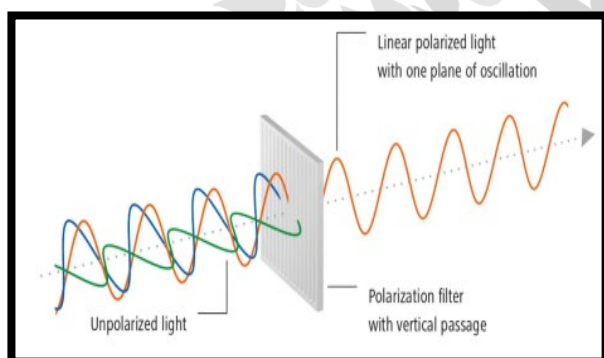
Diffraction of Light

Diffraction of light is the bending of light waves around the edges of an obstacle or aperture when the size of the obstacle/aperture is comparable to the wavelength of light.

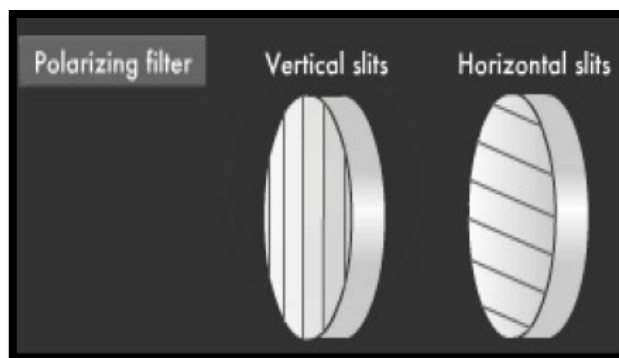
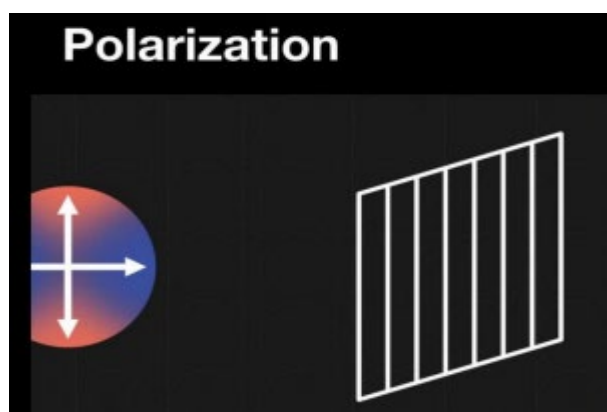
- More pronounced for longer wavelength
- Diffraction of light is the phenomenon when a ray of light gets bent and spreads out when it deviates from its path while passing through the edges of an obstacle or a hole
- This occurs when the wavelength of the light is approximately equal to the size of the obstruction or hole. Diffraction also illuminates the area that would normally be in the shadow area.

Polarization of Light

Polarization of Light is the phenomenon in which the vibrations of light waves are restricted to a particular direction (plane) perpendicular to the direction of propagation.



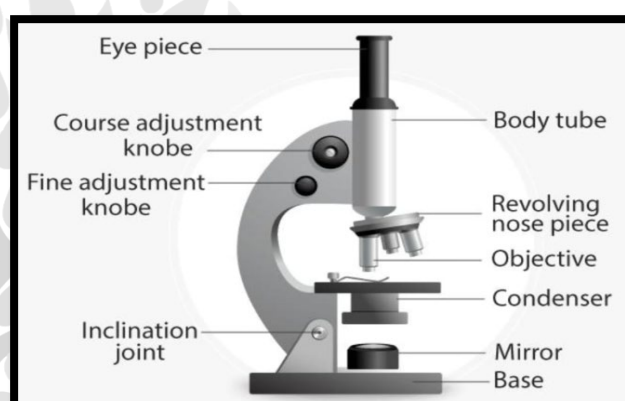
- Only transverse waves can be polarized
- Used in sunglasses, 3D movies, stress analysis
- After polarization, these vibrations are confined in a single plane, creating polarized light.



Optical Instruments

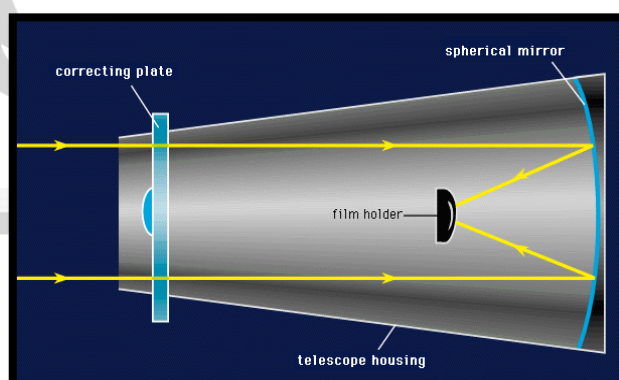
Microscope Compound

- Uses two convex lenses
- Produces enlarged, inverted image



Telescope (Astronomical)

A telescope is an optical instrument used to observe distant objects by collecting and magnifying light. It is widely used in astronomy to study stars, planets, and galaxies.



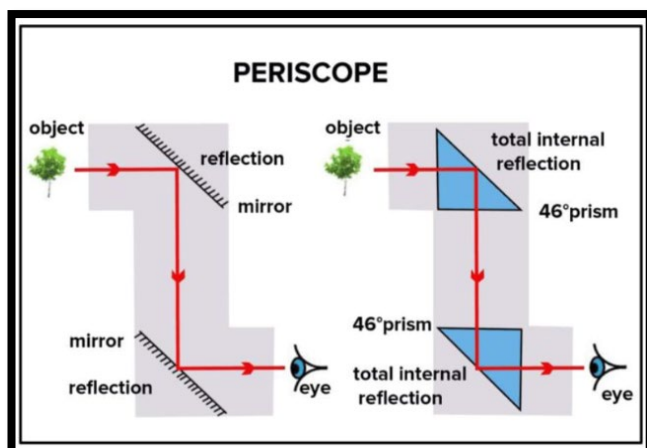
- **Objective** large focal length
- **Eyepiece:** small focal length
- Produces inverted image

Periscope

A periscope is an optical instrument used to see objects that are not in the direct line of sight,

especially from a hidden or enclosed position (like submarines or trenches).

- In periscope, two plane mirrors are inclined at 45° .



Kaleidoscope

A kaleidoscope is an optical instrument with two or more reflecting surfaces (or mirrors) tilted to each other at an angle, so that one or more objects on one end of these mirrors are shown as a symmetrical pattern when viewed from the other end, due to repeated reflection.

